INSTITUTE OF DISTANCE LEARNING

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI, GHANA

AE 464

CROP STORAGE AND STORAGE STRUCTURES

Lecture

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- **COURSE OBJECTIVE**
- Losses in store may outweigh the gains if knowledge on proper storage management is not acquired. The objective is to equip students with the principles of crop storage and the design of storage structure.

- **COURSE OUTLINE**
- **Introductory Concepts**
  - Reasons for crop storage
  - Physical factors of food deterioration in storage
  - Essential factors in crop storage
  - Storage environmental requirement

- **Durable crop storage**
  - Principles of durable crop storage
  - Storage types and their designs
    - Bulk storage vrs bag storage
    - Traditional storage structures
    - Conventional storage structures
    - Forces on silo walls
    - Janssen and Rankine's equations
    - Ware house storage

- **Aeration of produce**
  - Aeration systems- Natural and forced
- **Perishable crop storage**
  - Precooling
  - Cold storage systems
  - Refrigerated systems
  - Modified atmosphere
  - Controlled atmosphere
  - Hydro-cooling
- **Fruits and vegetables**
- **Tropical roots and tubers**

Reading Material

- Adel A. Kader (Ed.) 2010. Postharvest Technology of Horticultural crops.(Third edition). University of California, Agricultural and Natural Resources, Publication 3311
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- FAO (1985). Prevention of Post-Harvest Food Losses: A Training Manual. Rome UNFAO. 12Opp.
- David Bartecchi (2011). Crop Drying, Preservation, and Storage. Appropriate Technology Sourcebook

- Lisa Kitinoja and Adel A. Kader (2003) Small-Scale Postharvest Handling Practices: A Manual for Horticultural Crops (4th Edition). Postharvest Horticulture Series No. 8E Revised edition.
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Introduction

- **Why storage of Agricultural Produce?**
- Crop storage is a subject of particular relevance at a time when so many countries are struggling to overcome the effects of natural disaster. Whether the news is of drought or flooding, intense cold, or storm force winds, farmers often lose at least one season's crop production and many livestock.

Durable Crop Storage





The security of your grain is as good as your storage structure and the quality of its management

LECTURER:
Prof. A. Bart-Plange

The importance of storage

- They help to ensure household and community food security until the next harvest and commodities for sale can be held back so that farmers can avoid being forced to sell at low prices in the glut that often follows a harvest.

- While considerable losses can occur in the field, both before and during harvest, the greatest losses usually occur during storage.

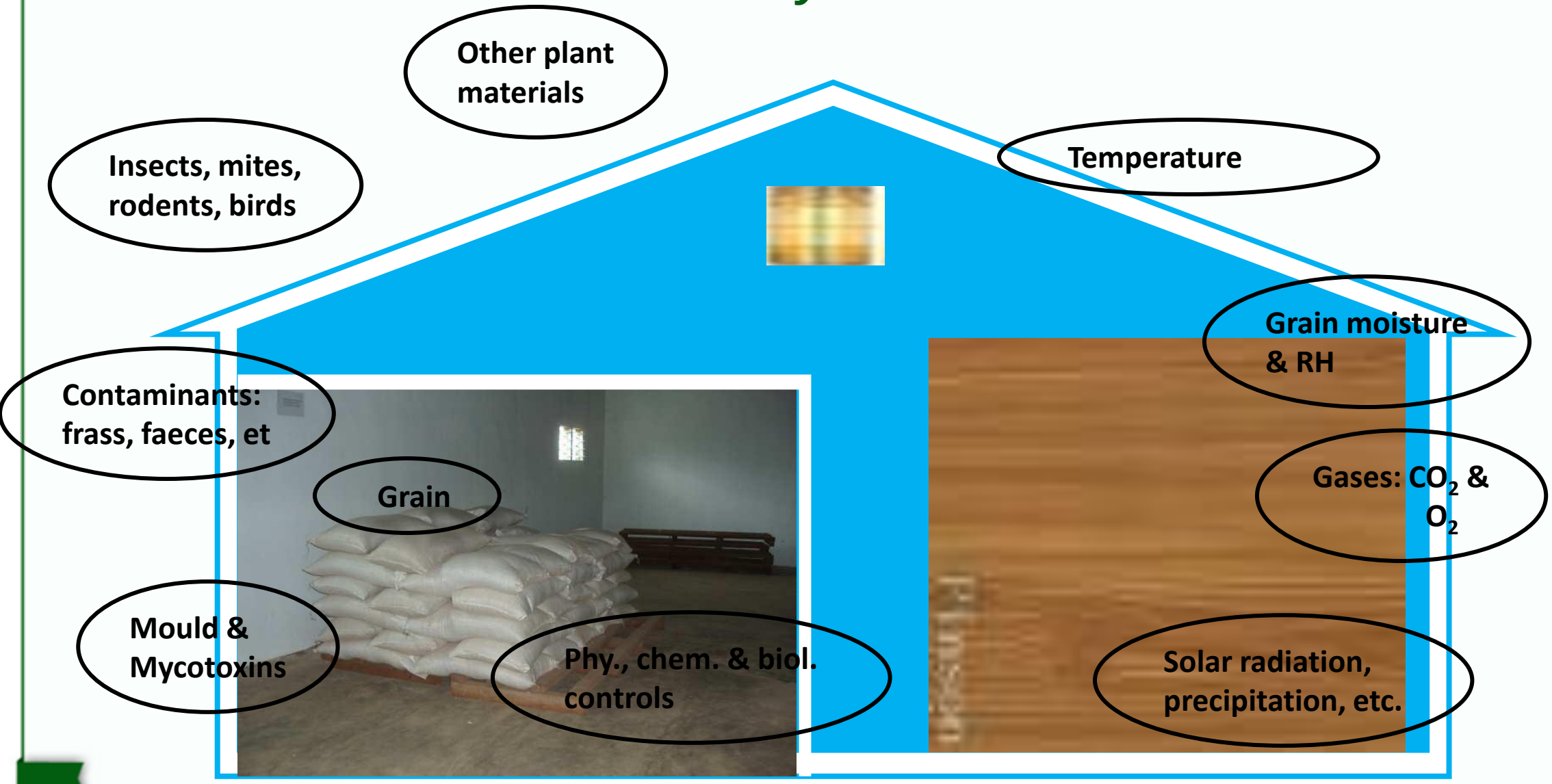
The basic objective of good storage

- Creation of environmental conditions which protect the product and maintain its quality and its quantity, therefore reducing product loss and financial loss.

Storage is thus useful for the following reasons:

- For future use
- To prevent waste
- For higher prices
- Taste and quality enhancement
- For planting materials for the next season
- Lack of ready market
- Means of savings
- Cultural practices
- Inaccessible roads
- Research/ germplasm (collection of genetic resources for an organism)

The Stored Grain Ecosystem



Basic Concepts of Stored Grain Management 1

- **Cereal grains and oilseeds are hygroscopic.** They lose moisture when exposed to dry air or gain moisture when exposed to wet air until equilibrium is reached.

- The weather, primarily the air vapour mixture properties determine strategies for drying and storing a particular grain type.

Basic Concepts of Stored Grain Management 2

- Grain is a biologically active
- Special care needed to prevent quality losses
- Stored grain is a man-made ecosystem which attracts pests and rodents which can reduce quality.
- Exclude and manage pests
- Grain quality WILL NOT IMPROVE during storage
- Important 1st step: understand the history and initial quality of stored grain

Important parameters of air for storage

- – **Dry Bulb Temperature:** is the air temperature measured by using a thermometer or temperature sensor.

- – **Relative Humidity (RH**): for a given temperature, the RH is the ratio of the vapour pressure of the water vapour contained in an air-vapour mixture to the vapour pressure of an air-vapour mixture that is completely saturated (100% RH).

Important parameters of air for storage

- **Equilibrium Moisture Content (EMC):** for a grain is the moisture content at which the grain will attain after being exposed to a given ambient condition. The grain equilibrates to the ambient relative humidity at the given ambient temperature.

Factors for food deterioration in storage

- **Physical Factors**
- Moisture content of the product (humidity)
- Temperature – grain temperature and ambient temperature

- **Biological factors**
- Vertebrates – rodents, birds
- Arthropods - insects
- Micro fauna – bacteria, moulds, etc

- **Technical factors**
- Storage conditions
- State of the product

- **Good storage thus involves controlling the following key factors within the storage environment:**

- Temperature
- Moisture
- Pests
- Hygiene

Temperature

- The temperature within a store is affected by the:

- sun,
- cooling effect of radiation from the store,
- outside air temperatures,
- heat generated by the respiration of both the food in store and any insects present.

Temperature

- Incorrect temperature control can result in physical damage.
- At high temperatures, fats in products may melt, while at low temperatures, sugar in sweet foods may crystallize.
- Chilling injury may occur. Considerable tissue disruption may occur when foods are frozen.
- Freezing also destroys the stability of emulsions, and foods such as milk and cheese may show a marked separation of fat upon thawing.

Moisture

- All micro-organisms, including moulds, require moisture to survive and multiply.

- All materials that have been dried will try to come back into equilibrium with the climate around them and in tropical countries this usually means absorbing moisture.

- The moisture level below which micro-organisms cannot grow is referred to as the safe moisture content.

Safe moisture content of stored products for long storage (average)

Product	Safe moisture content%
Maize	13
Millet	13
Sorghum	13
Rice	13
Cowpea	15
Groundnut	7
Cocoa	7
Palm kernel	5
Coffee	13

Moisture

- The siting and the ventilation of the store are important. If the walls of a store are cooled below their dew point by low night temperatures, condensation can occur and increase the moisture in the layers of produce near the edge of the store.

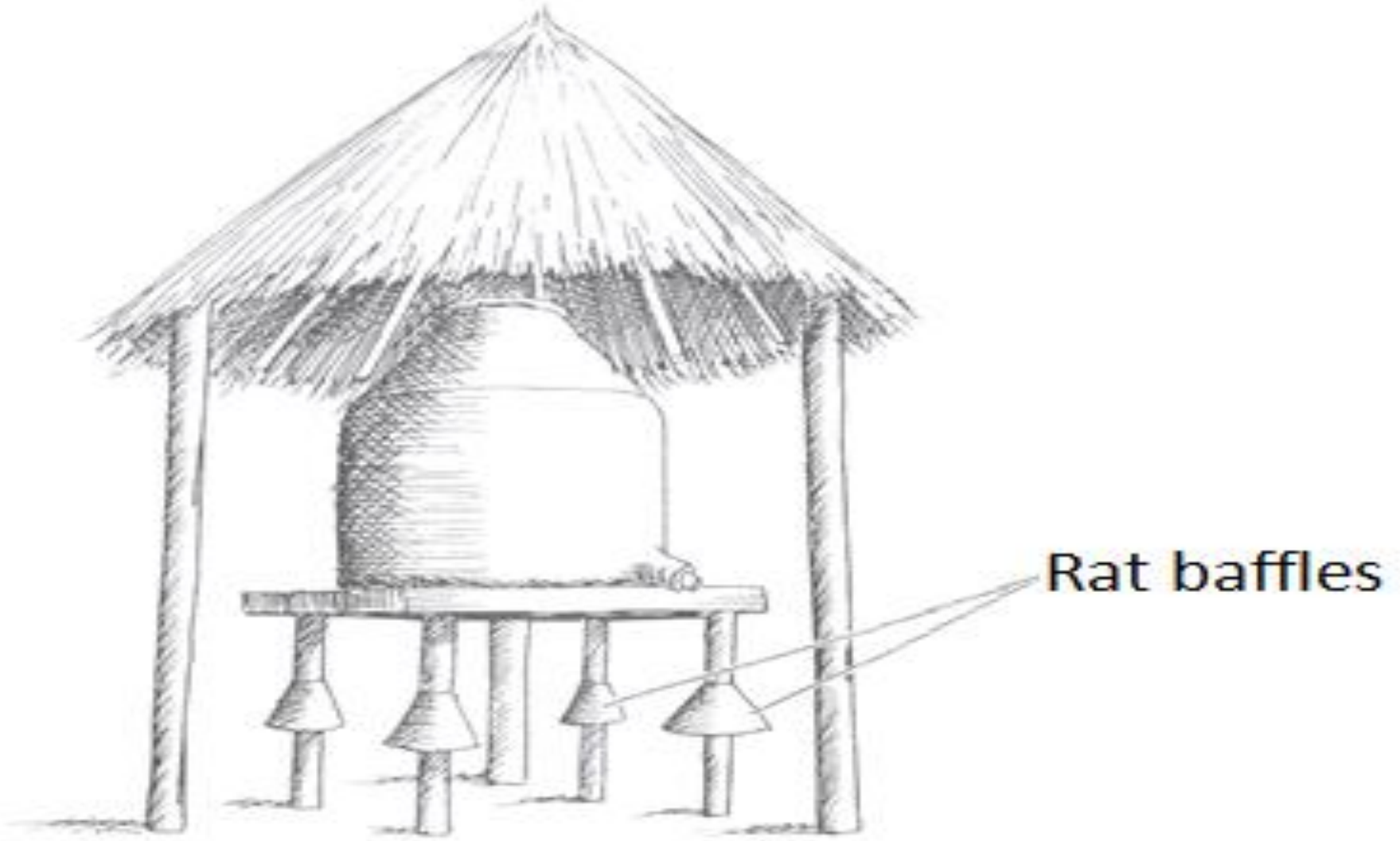
- It is important to remember that most stored food products are 'alive' and respiring, giving off moisture as well as heat.

Insects

- The main purpose of insect control is to prevent insects entering the storage containers.
- The containers should be closed tightly; baskets may be smeared with mud to seal them.
- Access to the store by insects such as ants and termites can be prevented by building the store on poles which are treated with coal tar, green camphor oil, or covered with fat or waste oil.
- Surrounding the supporting poles with a layer of sieved wood ashes, and sinking the poles into ground soaked with waste oil, will also deter insects.

PEST

- **Keeping pests out of the store**
- **Mammals**
- The most serious pests for store owners are rodents.
- All holes and entrances to the store should be fitted with fine wire mesh.
- The building should, if possible, be on poles with rat baffles around them as shown in the diagram.
- If the building is situated under a tree, rats can readily gain access on to the roof.
- The ground around the store should be cleared and all sources of water denied to rodents.
- They can easily gnaw through wooden doors so tin plate should be fixed to the bottom 30cm of the door.
- Domestic animals such as goats and poultry can pose a problem and should be kept at a distance from the store by build-ing simple protective fences.



Food store showing rat baffles

2. DURABLE CROP STORAGE

- Durable crop storage depends on:

1. Type of crop
2. Nature of the crop at harvest
3. Total volume of crop
4. Storage requirement for the crop
5. Form in which the crop is to be stored
6. Cost of storage

2.1 PROPERTIES OF STORED PRODUCTS

- Bulk density – It varies with the moisture content
- Angle of repose – It increases with the moisture content
- Static coefficient of friction
- Particle density
- Porosity

- **2.2 TYPES OF DURABLE STORAGE**
- Bulk storage
- Sack/bag storage

4.1.3Advantages and disadvantages of conventional bag and bulk storage

Bags	Bulk
Flexibility of storage	Inflexible storage
Partly mechanisable	Mechanisable
Slow handling	Rapid handling
Considerable spillage	Little spillage
Low capital costs	High capital costs
High operating costs	Low operating costs
Easy inspection	Inspection more difficult

3.1 Bulk Storage

- Bulk storage can be grouped into traditional and conventional/ modern

- Bulk storage techniques:

  - Hermetically sealed storage
  - Non-hermetically sealed storage

Hermetic Storage

What is hermetic storage?:

a technique for storing grain in a sealed air-tight container causing the living organisms (grains and insects) to deplete the oxygen and increase carbon dioxide of the enclosed microenvironment by their respiration to levels that will cause insects and mould not to survive.

Mechanism of Oxygen Depletion in Hermetic Micro-environment



O_2 CO_2 O_2 CO_2 O_2 CO_2

At the beginning:
Aerobic

At the end:
Anaerobic

Hermetic Storage Systems



GrainPro Cocoon™



Purdue Triple Inner-Bag (PICS)

SuperGrainBag

GrainPro, Inc.



3.2 Traditional bulk storage methods/structures

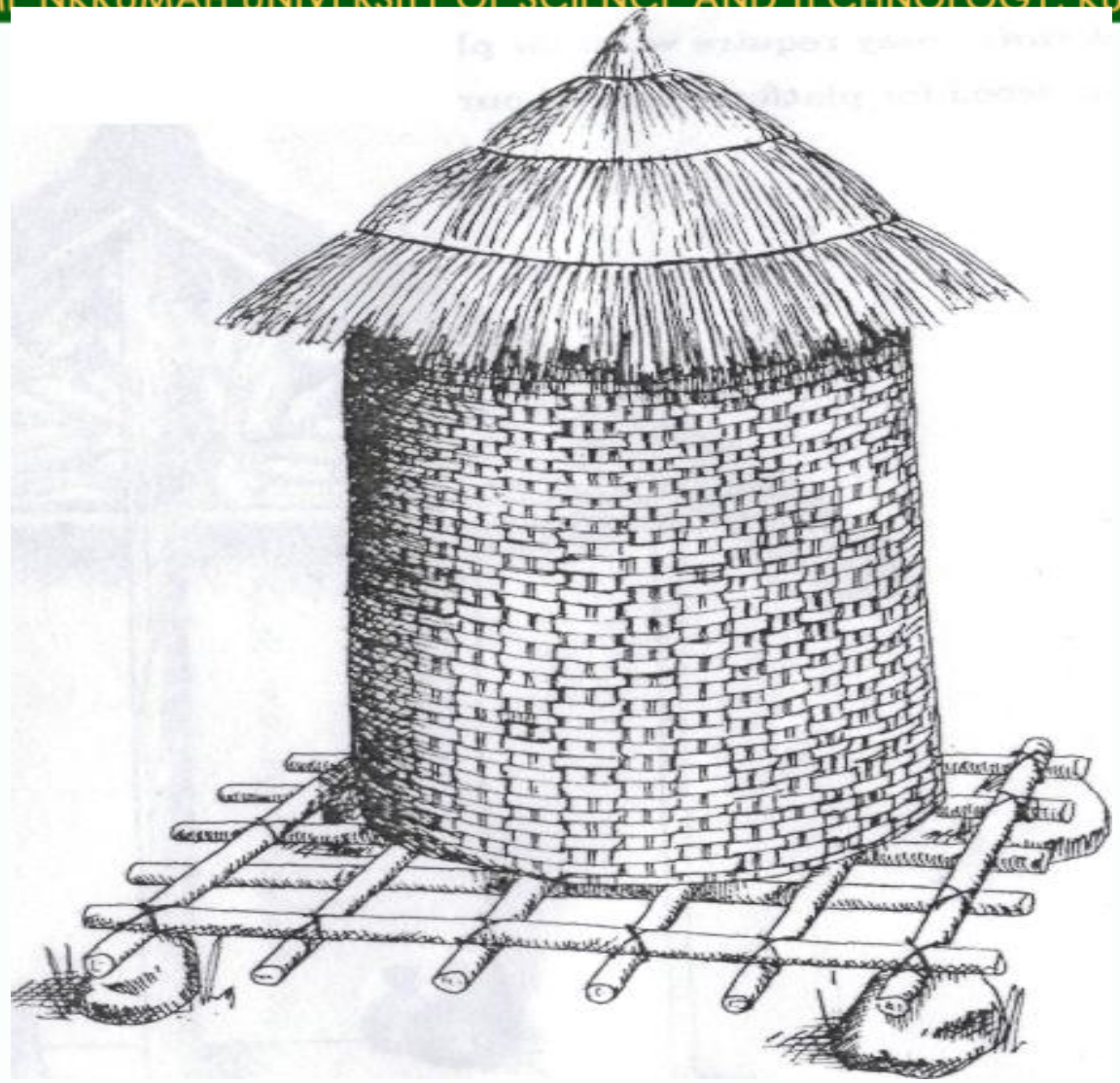
- ***3.2.1 Earthenware pots and gourds***
- In the tropics, earthenware pots and gourds (the hard dried outside skins of certain fruits and vegetables) are widely used for storing small quantities of food-stuffs.

- They need to be kept sheltered: above the kitchen where there are few insects is a good place.

- Such containers should be raised off the ground or put on plastic sheets to stop them absorbing moisture from below.

3.2.2 Baskets

- Baskets are widely used throughout the tropics for food storage.
- For certain products, good ventilation is required, so the baskets should not be placed too close together and if possible they should be raised off the ground.
- Woven baskets do not give much protection against insects.
- Baskets can be covered in plaster, mud or clay on both the inside and outside.



Basket silo on elevated platform for improved aeration

3.2.5Maize cribs

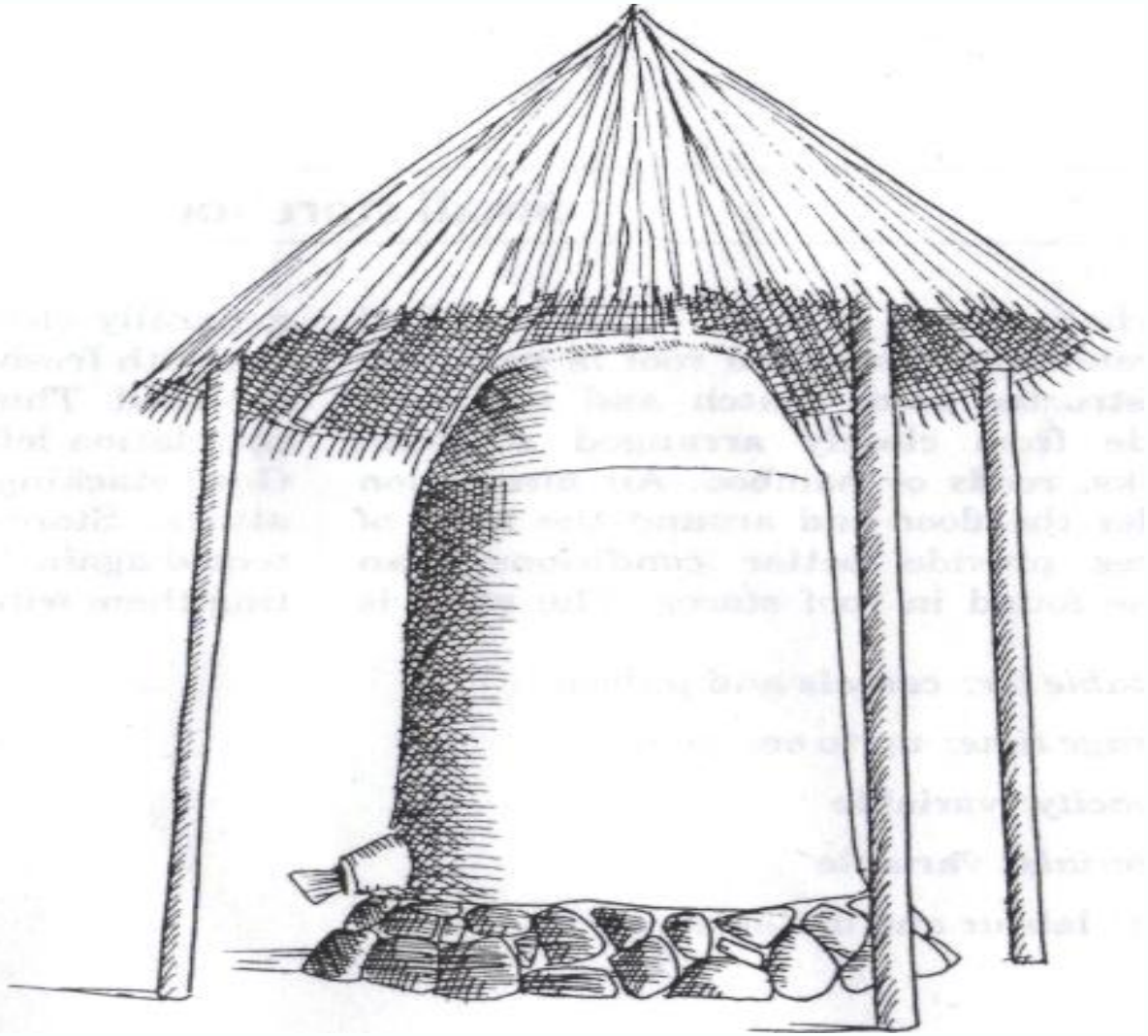
- Traditional maize cribs are suitable for the storage of maize cobs in semi-humid and dry tropics.
- The shape of the crib allows drying to proceed during storage using natural ventilation, provided that relative humidities are below 70 per cent.
- Cribs are preferably sited with their long side to the prevailing wind direction.
- Because of the narrow shape, usually 60-90cm in width, the drying process is better than in some traditional round cribs.

3.2.8Earth silos

- *Suitable for:* pulses and cereals

- *Storage time:* up to one year
- *Capacity:* variable

- *Materials:* earth, straw
- *Cost: low (*labour cost)



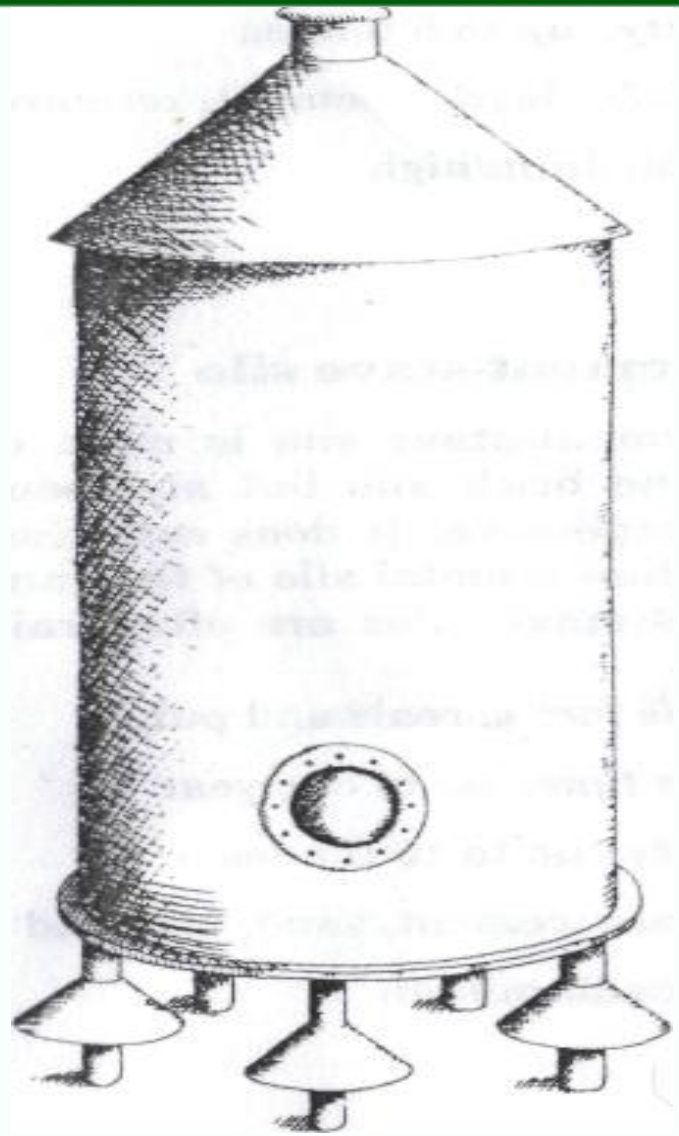
Earth silo

3.3 IMPROVED TRADITIONAL STORAGE STRUCTURES

- ***3.3.1 Plastic bags***
- These are widely used for storage in the humid and dry tropics.
- The product has to be dried well because, during storage, further drying is impossible as there is no air circulation.
- When plastic bags are well closed, airtight storage results, with all its advantages and disadvantages.

3.3.3Metal silos

- *Suitable for:* cereals and pulses
- *Storage time:* approximately one year
- *Capacity:* up to 5 tonnes
- *Materials:* sheet metal ;
- *Cost: m*edium/high,



Metal silo of 3-tonne capacity

4. MODERN OR CONVENTIONAL STORAGE STRUCTURES

- **4.1 Bulk storage structures**

➢Conventional structures:

- ✓Reinforced concrete silo
- ✓Corrugated steel silo
- ✓Brick wall silos
- ✓Plywood silos

4.2 Silo design considerations

- There are two main considerations in the design of silos:

  - Functional consideration
  - Structural considerations

- **4.2.1 Functional consideration**
- Provision of adequate volume
- Proper protection of the product to be stored
- Satisfactory method of loading and discharging of the product

4.2.2 Structural Considerations

- Applicable loads
- Materials used for construction of the silo
- Dead loads of material
- Live loads
  - From the stored products
  - Wind loads
  - Thermal loads
  - Impact and pressures

4.6 VOLUME REQUIREMENT FOR SILOS

- The volume requirement for a silo is determined by:
- Quantity of grain to be stored
- Type of grain to be stored
- Required cross section of the silo

- The parameters considered include:
  - Total weight of the product
  - Angle of repose
  - Cross section of the silo

Procedure for determining the volume requirement of a silo

1. Compute the mass of the product in kg. W_T
2. Determine the volume of the product. $V_T = W_T/\rho$
3. Consider the type of silo required either a shallow or deep silo.
4. Determine the equivalent diameter of the silo by relation between the volume of the products and the cross sectional area.
5. Use the equivalent diameter (D)of the silo to determine the height of the silo required.

For shallow silos, $(h) \leq D \tan (45^0+\theta/2)$

$$V_T = W_T/\rho$$

$$V_T = \frac{\pi D^2 h}{4} = W_T/\rho$$

Replacing h in equation above,

$$W_T/\rho = \frac{\pi D^3 \tan\left(45+\frac{\theta}{2}\right)}{4}$$

$$D^3 = \frac{4W_T}{\rho\pi \tan\left(45+\theta/2\right)}$$

EXAMPLE

- A farmer wants to select a silo which would be used to store 100 tons of maize and 80 tons of soya bean. Consider the silo as a shallow silo.

Grain	Density kg/m^3	Angle of repose
Maize	720	27
Soyabean	770	24

Solution

W_T= 100 tonnes = 100 x 10^3 kg

V_T = 138.89m^3

(h) ≤ D tan (45^0+θ/2), h is the height of grain in the silo

h ≤ 1.63D

$$V_T = \frac{\pi D^2 h}{4}$$

$$V_T = \frac{\pi D^2 (1.63D)}{4}$$

$$D = 4.8m$$

$$h \leq 1.63D$$

$$h \leq 7.8m$$

6. VENTILATION/AERATION IN STORAGE

Aeration is done to:

- Remove generated water
- Remove generated heat
- Maintain uniform environmental conditions
- Aeration design consist of determining the volume of air to be delivered per tonne minute and then designing the fan inlets, motor and system controls to deliver the desired air.

Standards for Maize in Ghana

- The Ghana Standards describes three types of maize:
  - yellow, white and red maize.
- For a maize type, at least 95% of the kernels in a lot must be of that type.
- **Moisture content shall not exceed 13% (wet basis)**
- **Total aflatoxin level shall not exceed 15 µg/kg (15**

ppb).

- There are three grades according to:

- ***Presence of other grains,***
- ***Degree of physical defects,***
- ***Foreign matter and filth.***

Standards for Rice in Ghana
GS 765: 2003

This standard applies to husked rice, milled rice, and parboiled

rice, all for direct human consumption:

- **Moisture content of 13% (wet basis) maximum.**
- **Total aflatoxin level:** shall not exceed 15 µg/kg (15 ppb).
- **Heavy metals**: Product must be free of heavy metals in amounts which may represent a hazard to human health.

- **Pesticide residues:** must be within maximum limits established by Codex Alimentarius Commission for rice.

Factors that affect Grain Spoilage in Storage

- Grain deterioration in a stored grain ecosystem depends on
- **Stored grain moisture**
- **Stored grain environment**
- – Temperature and Relative Humidity (RH)
- – CO_2 and O_2
- **Initial storage mould infestation**
- **Initial insect pest infestation and insect pest type**
- **Grain quality (presences of fines/intact kernels)**
- **Sanitation and storage structure**
- **Stored grain management**

The Concept of Grain Storability

- Rate of grain dry matter loss (quality and grade loss) due to growth of grain storage fungi/insects.
- Storage time to 0.5% dry matter loss is called the allowable storage time (AST).
- Grain storability is affected by prior post-harvest handling, kernel damage, storage conditions (moisture and temperature) and other factors.

Storage Life of Grain

°C	Maximum storage time (months) for corn and soybean*						
	Corn temperature ° F	Moisture Content Corn *(top %)*, Soybean *(bottom%)*					
		13%, 11%	14%, 12%	15%, 13%	16%, 14%	17%, 15%	18%, 16%	24% N/A
5	40	150	61	29.0	15.0	9.4	6.1	1.3
10	50	84	34	16.0	8.9	5.3	3.4	0.5
16	60	47	19	9.2	5.0	3.0	1.9	0.3
21	70	26	11	5.2	2.8	1.7	1.1	0.2
27	80	15	6	2.9	1.6	0.9	0.9	0.06

*Based on 0.5% maximum dry matter loss—calculated on the basis of USDA research at Iowa State University. Corresponds to one grade number loss; 2-3% points in damaged seeds. Soybean approximated at 2% lower moisture than corn.

Losses caused by mould growing on grain

- Decreased germination
- Discoloration of the germ
- Heating or mustiness
- Decrease in nutrient value
- Mycotoxins (***such as aflatoxin)*** *can be* produced by some moulds
- Loss of weight



Grain Losses Caused by Insects

Insect damage in a stored grain ecosystem depends on

- – Initial insect pest infestation
- – Grain nutritional value
- – Grain moisture
- – Moulds
- – Presence of predatory insect
- – Stored grain environment
- Temperature and RH
- **CO_2** and **O_2**

Thank You

For any concerns, please contact
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